

D&D; 5e

Encounter Builder

& Combat Tracker

A browser-based tool for Dungeon Masters to plan balanced encounters, manage party composition, and run turn-by-turn combat — all offline, no server required.

■ CR Budget System	■ Initiative Tracker	■ Habitat Filtering
■ Character Import	■ Monster Stat Blocks	■ Condition Manager
■ Custom Monsters	■ HP & AC Tracking	■ Combat Log

Tech Stack

HTML / CSS / Vanilla JS
localStorage (cross-page state)

Python http.server (local)
reportlab (this PDF)

open5e REST API v2
No build tools · No frameworks

ENCOUNTER BUILDER

D&D 5e · Challenge Rating System

PARTY SETUP

Fighter

5

+

-

Cleric

5

+

-

Rogue

5

+

-

+ Add Party Member Manually

IMPORT FROM CHARACTERS

Locran Darter

Lvl 3 Warlock

+

Trent Anteson

Lvl 20 Assassin

+

benson

Lvl 11 Ranger

+

samson

Lvl 10 Wizard

+

Members: 3

Total Level: 15

+ ADD PARTY TO ENCOUNTER

DIFFICULTY

Trial

1%

Easy

1%

Moderate

1%

Hard

1%

Deadly

1%

7.5

CR Threshold

ENCOUNTER PROGRESS

0 CR used

750 CR remaining

75 CR

ADD MONSTER

NAME

TYPE

HABITAT

CR MIN

CR MAX

Dragon, Goblin...

All

All

All

All

SEARCH MONSTERS

+ Add Custom Monster

MONSTERS IN ENCOUNTER

No monsters added yet

CLEAR ALL

EXPORT

+ START COMBAT

Encounter Builder — main view

Page 1

Encounter Builder

Party Setup

- Add members manually or import saved characters
- Per-member level control (1–20)
- Characters loaded from the Character Builder via shared my_characters/ folder
- Add Party to Encounter button places PCs directly into the monster list

Difficulty System

- Five tiers: Trivial · Easy · Moderate · Hard · Deadly
- CR threshold calculated from total party level
- Live CR budget bar shows usage with colour feedback
- Over-budget warning with exact percentage

Monster Search

- Filters: Name, Type, Habitat, CR Min/Max
- Powered by open5e API v2 (331 SRD 2024 monsters)
- Habitat data from D&D; 5e Monster Manual tables
- Coloured habitat chips on each stat card

Custom Monsters

- Add any monster not in the SRD
- Fields: Name, CR, HP, AC, Initiative Bonus, Type, Size
- Stats carried through to the combat tracker
- Toggle panel keeps the UI clean

ENCOUNTER BUILDER

D&D 5e · Challenge Rating System

PARTY SETUP

Fighter

5

+

Cleric

5

+

Rogue

5

+

+ Add Party Member Manually

IMPORT FROM CHARACTERS

Locran Darter

Lvl 5 Warlock

+

Trent Anteson

Lvl 20 Assassin

+

Denson

Lvl 11 Ranger

+

Samson

Lvl 19 Wizard

+

Members: 3

Total Level: 15

+ ADD PARTY TO ENCOUNTER

DIFFICULTY

Trivial

Easy

Moderate

Hard

Deadly

7.5

CR Threshold

ENCOUNTER PROGRESS

0 CR used

7.50 CR remaining

/ 7.5 CR

ADD MONSTER

NAME

TYPE

HABITAT

CR MIN

CR MAX

nothin

All

All

All

All

SEARCH MONSTERS

+ Add Custom Monster

5 monsters found

Goblin Minion

Small Fey

CR 1/8

Goblin Warrior

Small Fey

CR 1/4

Hobgoblin Warrior

Medium Fey

CR 1/2

Goblin Boss

Small Fey

CR 1

Hobgoblin Captain

Medium Fey

CR 3

Loading...

MONSTERS IN ENCOUNTER

No monsters added yet

CLEAR ALL

EXPORT

+ START COMBAT

Monster search results with full stat card — Goblin Warrior

Page 2

Encounter in Progress

[illegible]

Party members (blue) and monsters listed side-by-side with CR budget tracking. Custom monster form expanded. Clickable monster rows re-render the cached stat card without re-fetching.

Data Architecture

File	Purpose
<code>data/habitats.js</code>	window.HABITAT_DATA — 331 creature habitat mappings loaded via <code><script></code> tag (CORS-safe)
<code>my_characters/characters.js</code>	window.SAVED_CHARACTERS — party members exported from the Character Builder
<code>index.html</code>	Single-page app: party setup, monster search, CR budget, encounter list
<code>combat.html</code>	Combat tracker: initiative phase → combat phase with full stat blocks
<code>styles.css</code>	Dark theme for encounter builder (CSS variables)
<code>combat.css</code>	Dark theme for combat tracker

Combat Tracker

Initiative Phase

- Card per combatant with editable HP, AC, and Initiative fields
- Real stats pre-filled from API data or character JSON
- Roll All Creatures button auto-rolls d20 + bonus for all monsters
- Animated number pop-up on each roll
- PCs shown with blue tint; monsters with CR badge

Turn Order

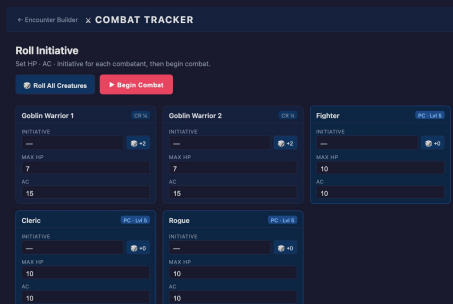
- Combatants sorted high-to-low initiative, ties broken by bonus
- Scrollable token strip across the top of the combat view
- Active combatant glows amber; click any token to jump
- Defeated monsters shown at reduced opacity

Active Combatant Panel

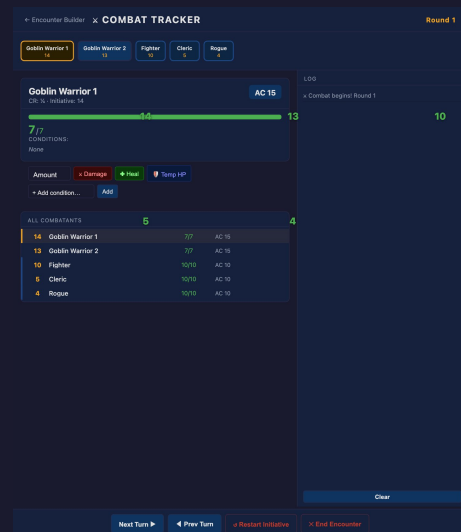
- HP bar with colour feedback (green / yellow / red)
- Damage, Heal, and Temp HP controls
- Condition tags with colour-coded removal
- Death save tracker for downed PCs

Stat Block Viewer

- Click clipboard icon on any combatant row
- Full stat block rendered from cached API data — no re-fetch
- Monster: ability scores, saves, resistances, traits, all actions
- PC: ability scores, spell slots, passive perception, speed



Initiative phase — cards with pre-filled stats



Combat phase — turn strip, active panel & log

Built entirely with vanilla HTML, CSS, and JavaScript — no frameworks, no bundler, no backend. Runs offline as a local file served by Python's built-in http.server.